

1. Specification Intelligence Page

Input Guardrails

- **File Upload Sanitizer:** Validates file size ($\leq 25 \text{ MB}$), allowed extensions (.pdf, .docx, .txt), and magic-byte binary signatures.
Protects From: Executable malware uploads, file extension spoofing, and oversized file memory crashes.
- **Document Text Security Scanner (Powered by Llama Prompt Guard):** Scans specification text chunks using Llama Prompt Guard and regex injection patterns.
Protects From: Indirect prompt injection attacks embedded inside PDF/DOCX spec files.

Output Guardrails

- **Pydantic Schema Validator:** Enforces strict JSON schema structure on extracted requirement outputs.
Protects From: Malformed JSON, missing fields, or broken data contracts.
- **Knowledge Graph Hierarchy Validator:** Verifies directed relationships linking 3GPP references (3GPP TS 38.331) to child clauses (Clause 5.3.1).
Protects From: Disconnected, broken, or orphaned requirement knowledge graphs.
- **Source Clause Traceability Check:** Ensures extracted requirements map back to valid source clause files.
Protects From: Untraceable requirements and missing specification clause mappings.
- **NLI Spec Groundedness Check:** Uses a local DeBERTa NLI cross-encoder model to verify extracted claims against original 3GPP source text.
Protects From: LLM hallucinations, invented requirements, and false specification claims.

2. Test Script Generator Page

Input Guardrails

- **Prompt Injection Scanner (Powered by Llama Prompt Guard):** Scans user prompts using Llama Prompt Guard
Protects From: Direct jailbreaks, system prompt overrides
- **Telecom Scope Validation:** Verifies template topic relevance against telecom and testing domain rules.
Protects From: Saving off-topic or unauthorized custom templates.

- **Refine Prompt Scope Check:** Validates follow-up user commands against active session context.
Protects From: Session hijacking and prompt drift during interactive code refinement.

Output Guardrails

- **Python Syntax AST Check:** Parses generated test script code using Python AST.
Protects From: Syntax errors, broken code, and non-executable Python scripts.
- **3GPP Clause Traceability Validator:** Scans generated code for mandatory 3GPP clause reference comments.
Protects From: Non-compliant test scripts lacking spec traceability.
- **3GPP Intent Coverage Graph Check:** Compares generated test cases against mandatory 3GPP protocol intents.
Protects From: Incomplete test suites that miss critical spec requirements.
- **Scenario Signaling Message Coverage:** Validates required signaling message sequences (Registration Request/Accept, PDU Session Setup, Handover).
Protects From: Invalid protocol signaling flows and misclassified test scenarios.

3. Bug Discovery & Log RCA Page

Input Guardrails

- **Telecom Domain Checker:** Fingerprints log headers and messages against 5G gNB runtime and build log patterns.
Protects From: Analyzing non-telecom logs (OS event logs, web server logs, database queries).
- **Historical Pattern Matcher:** Compares log execution milestone sequences against past completed RCAs.
Protects From: Running automated RCA on unknown or unsupported log failure patterns without human review.
- **Log Data Quality & Completeness Check:** Detects mid-line truncations, missing timestamps, blank gaps, and incomplete event tails.
Protects From: Performing RCA on corrupt, truncated, or incomplete log captures.
- **Scenario Relevance Check:** Verifies expected signaling events for reported test scenarios.
Protects From: Mismatched log analysis (e.g., analyzing a Registration log under a

Handover scenario).

- **Log Security Scan (Powered by Llama Prompt Guard):** Scans log text for embedded injection attacks using **Llama Prompt Guard**.

Protects From: Malicious logs containing prompt injection payloads aimed at hijacking the downstream LLM log analyzer.

Output Guardrails

- **Context Confidence Scorecard:** Aggregates diagnostic confidence scores against historical RCA database benchmarks.

Protects From: Low-confidence or inaccurate root cause diagnosis reports.

4. Prompt Studio Page

Input Guardrails

- **Template Security Scanner (Powered by Llama Prompt Guard):** Scans user templates, placeholders, and reference documents using **Llama Prompt Guard** and regex injection rules.

Protects From: System prompt overrides, command execution payloads in template variables (e.g., `${var}`), and malicious attached context files.

- **Telecom Scope Validation:** Verifies template topic relevance against telecom and testing domain rules.

Protects From: Saving off-topic or unauthorized custom templates.